IRRIGATION WATER MANAGEMENT · NOTE WM05

Farm Irrigation Structures — Conveyance, Erosion-Control and Water-Control Structures

Course topics: IDE5.5 L7, IDE-501 T8

Farm irrigation structures convey and control water on the farm. The distribution system must carry the stream safely from the source to the furrow, basin or border, give control, and keep labour to a practical minimum — good structures are an essential part of an efficient layout.

Conveyance structures — crossing roads, hillsides and depressions

Structure	Use and design points
Flume	Overhead channel of wood, metal or concrete on pillars or fixed to bridges, carrying water across gullies, ravines and depressions. Alternate wetting and drying make wooden flumes hard to keep watertight. Give it enough slope for a higher velocity than earth channels so the section can be smaller. Supports: creosote-treated timber, painted steel, concrete or masonry
Pipe flume	Steel or concrete pipe on pillars, set below the water surface so it flows full; a steeper pipe grade increases capacity
Culvert	Where a channel crosses a road, ridge, depression or low stream — enough when the road fill is high and the channel grade rests on the field surface. Corrugated metal, concrete or steel pipe; keep the velocity high (no silting, smaller pipe); finish the outlet with a headwall against traffic damage and erosion; minimum earth cover 30 cm (corrugated metal) and 20 cm (concrete); stilling basin at the outlet
Inverted siphon	Where the road is too close to the surface for a culvert or the channel is higher than the road: inlet and outlet tanks joined by a pipe near their bottoms (concrete, corrugated metal, steel or vitrified tile). Check gates on the inlet tank control the upstream level; the tanks act as stilling basins but collect silt, so give access for cleaning (common size 60 cm × 60 cm). Prefer larger pipes for easy cleaning and keep enough earth cover against traffic

Erosion-control structures — channels on steep hillsides

Uncontrolled flow down a steep slope cuts gullies. Use drop structures, chute spillways and drop-inlet structures.

- **Drop structures** lower water from a higher to a lower level and dissipate its energy; protect the downstream side against undermining. Small drops: a simple stone or concrete apron, even a discarded drum (Fig. 7.22); larger drops: masonry or concrete (Figs 7.23, 7.24).
- Eddies can slough the ditch banks at the lower end of drops, at pipe drop inlets and at sharp bends — protect these with rock riprap (lined channels seldom erode).
- **Pipe drop structures** carry water down through a pipe where an open drop would disturb an existing bund or dam: vitrified sewer, concrete, corrugated metal or PVC pipe; a watertight lid at the inlet doubles as a check gate; a masonry or concrete apron round the inlet stops seepage; sewer pipes use large-radius bends; a stilling basin at the outlet dissipates energy.
- **Chutes** — paved ditches of concrete, stone or brick that carry water down slopes too steep for closely spaced drops; build them for high velocity and slow the water in a stilling basin before it

enters the lower channel.

Water-control structures

They make water application easy and accurate, cut labour and make the best use of the supply.

Structure	Description
Diversion box	Splits a main channel into two or more branches; brick, stone masonry or concrete, with metal or wooden check gates sliding in grooves at the branch entrances (proportional diversion boxes; two-way diversion-and-drop structures)
Check	A small dam across a channel to hold the water level for irrigation, with metal slide gates or wooden shutters; in earth channels add an apron, preferably concrete, against scour from the overpour
Portable check dam	Canvas or sheet-metal dams (or masonry checks with slide gates) to check or raise the water level in unlined channels: bury the canvas bottom and sides in the banks and bed for a seal; drive the sheet metal into them
Turnout	Takes water from a lateral into a field ditch or straight into the field; portable or built-in, gated; in lined channels build turnouts with the lining
Siphon tube	Plastic, rubber or metal (aluminium or sheet iron) tube carrying water over the ditch bank into a field or furrow; easier to operate than spiles. Flow is controlled by the number of tubes or the height of their outlet ends
Spile / field pipe	Bamboo, sheet-iron or baked-clay pipe or a wooden lath box set in the ditch bank, taking water from a secondary ditch into the field; square wooden boxes and bamboo pipes are common

$$\text{Siphon tube: } Q = 0.65 \times 10^{-3}\, a\, \sqrt{(2\, g\, h)}$$

Q = discharge (L/s); a = internal area of the tube (cm²); g = 981 cm/s²; h = effective head (cm) — the difference between ditch and furrow water levels for a submerged outlet, or the height of ditch water above the centre of the outlet end for a free outlet.

> A 4 cm siphon (a = 12.57 cm²) under 15 cm effective head: $Q = 0.65 \times 10^{-3} \times 12.57 \times \sqrt{(2 \times 981 \times 15)} = 0.00817 \times 171.6 \approx 1.4$ L/s.

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. A structure carrying an irrigation channel over a gully on supports is a

(a) culvert (b) flume
(c) siphon (d) drop

Q2. When the channel water surface is higher than a road, the crossing used is

(a) a culvert (b) an inverted siphon
(c) a chute (d) a spile

Q3. Minimum earth cover over a corrugated-metal culvert pipe is

(a) 10 cm (b) 20 cm
(c) 30 cm (d) 60 cm

Q4. A paved ditch that carries water down a steep slope is a

(a) flume
(b) chute
(c) check
(d) turnout

Q5. Siphon-tube discharge coefficient (Q in L/s, a in cm², h in cm) is

(a) 0.61×10^{-3}
(b) 0.65×10^{-3}
(c) 0.0184
(d) 0.0138

Answers

Q1 — (b) flume
Q2 — (b) an inverted siphon
Q3 — (c) 30 cm
Q4 — (b) chute
Q5 — (b) 0.65×10^{-3}